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EchoNet++: A multilingual soccer match audio commentary dataset

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We present a robust, multilingual, and multistage audio analysis pipeline for automatic processing of professional soccer match broadcasts. Rather than proposing a new standalone ASR model, this work introduces a reproducible end-to-end system and systematic component-level analysis that reveal how preprocessing, segmentation, transcription, and translation jointly affect performance under real-world broadcast conditions. The pipeline performs audio extraction, denoising, speech segmentation, transcription, and translation in a unified framework, enabling scalable, language-agnostic audio understanding. Audio is extracted from match videos using FFmpeg, transformed into the frequency domain via FFT, and filtered with a Butterworth bandpass filter (300 Hz–7 kHz) to isolate speech-relevant frequencies. We apply deep learning-based denoising using Demucs and segment speech regions using Silero VAD, followed by categorization of commentator and spectator streams based on temporal and density cues. Each speech segment is transcribed using multiple Automatic Speech Recognition (ASR) models, including Whisper variants (Medium, Large, Turbo) and Insanely Fast Whisper, and non-English transcripts are translated into English. All results, including transcriptions, translations, and metadata (timestamps, segment labels, and language codes), are exported as match-structured JSON files. We evaluate our pipeline on full-match videos from Europe's top leagues, achieving low word error rates, accurate language detection, and consistent segmentation across diverse acoustic conditions. To support reproducible research and downstream tasks such as summarization and analytics, we will publicly release the dataset and processing pipeline.

Keywords Soccer broadcast commentary analysis, Multilingual audio processing, Automatic speech recognition, Voice activity detection, Audio denoising and segmentation

Modern sports analytics is rapidly evolving beyond traditional visual processing. While early approaches to soccer analysis focused solely on video-based tasks, such as player tracking [1–4], action spotting [5–7], or camera calibration [8], recent efforts have embraced multimodal integration, combining video, audio, and text streams to offer a more holistic understanding of complex match dynamics [5]. Alongside the commentator's voice, crowd reactions and ambient stadium sounds encapsulate critical contextual signals that complement visual analysis. Harnessing these multimodal auditory cues presents a valuable opportunity for a deeper understanding of game events, emotional shifts, and narrative structure.

Recent datasets, such as SoccerNet-Echoes [9], have taken a pioneering step by augmenting soccer videos with automatically transcribed audio using Automatic Speech Recognition (ASR) models, like OpenAI Whisper [10], and translating non-English content via Google Translate. These efforts allow textual enrichment for downstream tasks such as action spotting and summarization [11]. Despite growing interest in audio-based analysis, most existing approaches rely on surface-level ASR techniques with minimal preprocessing. However, they fail to handle the full complexity of broadcast audio, including overlapping speakers, unstructured crowd noise, domain-specific transcription errors, and the absence of robust segmentation strategies. The acoustic complexity of soccer broadcasts presents several challenges, including commentator speech often overlapping with crowd noise, background music, or environmental distortions; language shifts can occur within a single broadcast; and the use of low-resource languages or dialectal variations further compromises transcription accuracy. Moreover, traditional ASR pipelines tend to overlook key signal processing steps, such as denoising [12,13], filtering [14], speaker segmentation [15], and volume normalization, all of which are crucial for consistent performance across diverse recording conditions.

To address these challenges, we introduce a comprehensive and scalable audio analysis pipeline tailored for professional soccer match videos. Our system goes beyond simple transcription by implementing a multistage framework combining signal processing, speech isolation, multilingual transcription, and translation. It

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enables detailed semantic and structural exploration of match audio, supporting robust downstream analytics. This work is primarily a systems-oriented, learning-enabled signal processing study that investigates robust speech transcription in noisy sports broadcast environments. While we introduce a curated dataset to support reproducible evaluation, the dataset serves as an enabling resource rather than the sole contribution of this paper. This work presents several key contributions toward advancing systems-oriented, learning-enabled audio signal processing for sports analytics.

- We introduce EchoNet++, a new dataset with full metadata, made available as an open-access resource. To create it, we curated a subset of 500 .mkv clips from the SoccerPro dataset, comprising 20 full matches across six major leagues in three quality tiers (720p@50 FPS, 1080p@30 FPS, 1280p@30 FPS). These clips were augmented with automatically transcribed and translated audio commentaries from soccer game broadcasts in multiple languages.
- We develop EchoNet, a systems-oriented, learning-enabled signal processing pipeline for robust transcription of noisy, multilingual sports broadcast audio. The end-to-end framework integrates FFT-based spectral filtering, Butterworth bandpass filtering, deep learning-based denoising (Demucs), neural voice activity detection (Silero VAD), multilingual transcription using Whisper and Insanely Fast Whisper (IFW), and automatic translation to English.
- We introduce the Report Accuracy (RA) metric, defined as RA $\approx 100 - 0.95\cdot$ WER, to quantify end-to-end transcription-to-translation alignment. RA bridges low-level signal enhancement with high-level task accuracy, providing a single, interpretable score for practical evaluation and assessment. Applying the EchoNet preprocessing pipeline consistently improves RA across datasets and models. On EchoNet++, RA rises from 80.6% to 86.6% (Whisper(M)), 81.9% to 87.6% (Whisper(L)), 82.3% to 88.2% (Whisper(Turbo)), and 83.3% to 89.4% (IFW). Similarly, GOAL shows gains from $83.1 - -85.6\%$ to $88.1 - -91.2\%$, and SoccerNet-Echo from $81.8 - -84.3\%$ to $86.9 - -90.0\%$. These results demonstrate that EchoNet not only reduces transcription errors but also substantially enhances the reliability of downstream report generation.
- We demonstrate the utility of our pipeline in real-world scenarios, enabling applications such as multilingual commentary analysis, cross-lingual sentiment tracking, and automated highlight generation, thereby advancing the field of audio-visual sports analytics.

Our framework offers a robust, modular, and language-agnostic solution for semantic soccer audio analysis (see Fig. 1). It addresses practical deployment challenges while enabling high-quality downstream tasks such as commentator behavior analysis, automatic highlight generation, and multilingual game summarization, all grounded in reliable speech understanding from noisy, real-world inputs.

Related work

Recent advancements in audio-driven understanding of multimedia content have spurred a new wave of research in multimodal sports analytics. Traditionally, the focus has been on visual-based cues such as player tracking, action recognition, and dense video captioning[5,16,17]. However, visual-only methods often overlook vital contextual cues embedded in game audio, such as commentator emphasis, crowd reactions, or tonal

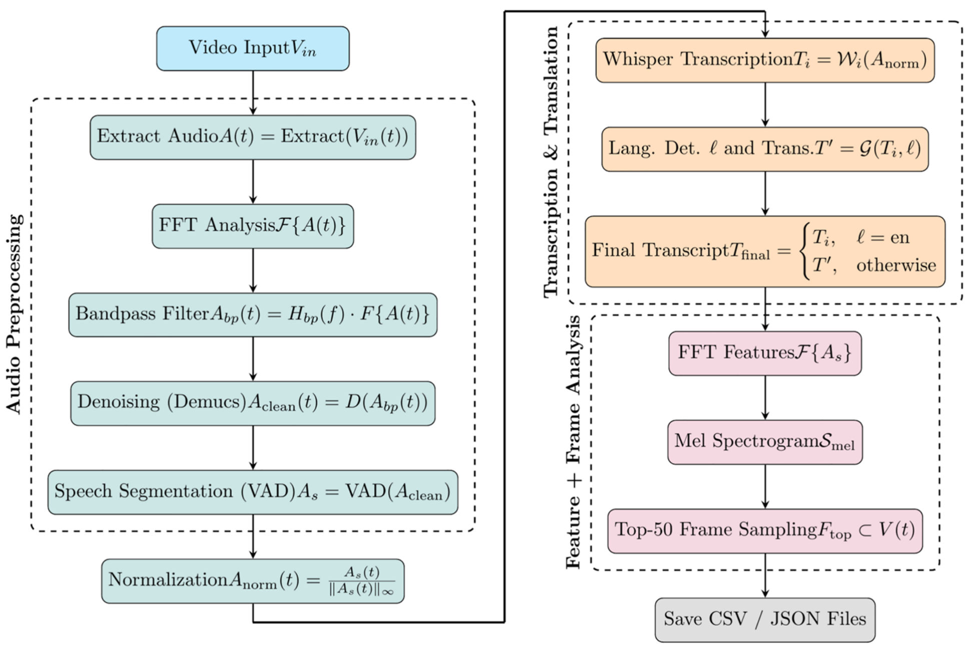

Fig. 1. Overview of the EchoNet audio processing pipeline, highlighting multilingual speech transcription, English translation, speech activity detection, and keyframe alignment for soccer broadcast analysis.

variations, which can enrich semantic understanding and improve downstream tasks like event detection and summarization [18,19]. This has motivated the integration of automatic speech recognition (ASR), voice activity detection (VAD) [20,21], and audio denoising technologies into sports analysis workflows [22]. Our work builds upon and significantly extends these foundations by introducing a high-fidelity pipeline tailored for broadcast soccer audio, emphasizing robust preprocessing, segmentation, multilingual transcription [23,24], and translation, all validated across diverse real-world league recordings.

Audio analysis in sports

The role of audio in sports understanding has evolved significantly, from detecting ambient noise to analysing structured commentator speech. Early works in audio-based event detection, such as those by Baillie et al. [25], explored the classification of crowd noise and referee whistles; however, they often lacked scalability and robustness in live broadcast scenarios. More recently, multimodal studies have demonstrated the value of combining visual and audio cues for tasks such as action spotting, highlight extraction, and dense captioning [5,17,23,26]. The SoccerNet-Echoes dataset represents a shift towards integrating commentator audio into structured analysis using ASR [9], though it primarily focuses on applying Whisper for transcription and Google Translate for language normalization.

In contrast, our pipeline addresses several gaps by incorporating advanced noise filtering, segment-level classification (crowd vs. commentary), and robust multilingual handling. Sun et al. [27] utilize commentator speech to enhance temporal segmentation of key moments, while Ikeda et al. [28] analyze Japanese baseball commentary using Whisper and ReazonSpeech models to overcome linguistic challenges. Our work generalizes this approach across multiple European leagues and languages, coupled with an open-source dataset release for community use.

Transcription models under noise

Large-scale ASR models have revolutionized transcription accuracy, particularly in noisy and multilingual environments. OpenAI's Whisper [10] is leading in benchmarks, being trained on 680k+ hours of multilingual audio. Its transformer-based encoder-decoder architecture enables robust transcription across varied accents and speech domains. However, inference costs remain high for large models. CTranslate2-based optimizations, such as Insanely-Fast-Whisper(IFW) [29], reduce latency significantly by quantizing the models and enabling FlashAttention2. Several studies have evaluated Whisper variants in sports domains where dense, overlapping speech and crowd noise present challenges. Azure ASR and Google ASR remain competitive in low-resource deployments [30] but lack the fine-tuning potential and multilingual depth of Whisper. Our work benchmarks multiple Whisper variants in realistic broadcast environments and demonstrates superior flexibility across latency accuracy trade-offs.

Voice activity detection and denoising

Segmenting meaningful speech content from noisy audio is a prerequisite for effective transcription. Traditional VAD techniques, often rule-based or energy thresholding methods, perform poorly under crowd interference and background music. Recent neural models like Silero VAD [15,31] offer real-time, low-latency VAD with high accuracy, even in complex auditory scenes.

We adopt Silero due to its efficiency and high recall in separating commentator segments from crowd reactions. Complementary to VAD, deep learning-based denoising has matured with models like Demucs [13] and its more recent iterations, which use convolutional encoder-decoder architectures to isolate vocals or suppress environmental noise. Demucs have been widely adopted in music source separation and are increasingly applied to noisy ASR preprocessing pipelines. Our use of Demucs in tandem with filtering (via Butterworth bandpass between 300Hz and 7kHz) ensures high signal clarity before transcription. While recent datasets like GOAL [32] and SoccerNet-Echoes[9] have introduced audio-based transcription and translation pipelines, their implementations often remain limited to baseline ASR models, without detailed preprocessing or robust segmentation strategies. Moreover, issues like overlapping speaker resolution, unstructured crowd filtering, and metadata-preserving storage formats are not comprehensively addressed. Our contribution fills this methodological gap by offering a modular, open-source pipeline that spans audio denoising, VAD-based segmentation, multilingual transcription with model benchmarking, and final-stage translation with structured JSON output validated across multilingual broadcasts and released as a reusable dataset.

Methodology

In this study, we utilize our SoccerPro [3] dataset, which was developed prior to our instance segmentation task. This dataset consists of a large collection of professional soccer match videos, containing 1, 495 .mkv files and totaling approximately 4.7 TB of data, though it has not been released publicly. The full dataset includes recordings of 47 complete matches with varied resolutions and frame rates. Each match has a consistent duration of 90 minutes. We curated a representative subset consisting of 500 .mkv video clips derived from 20 complete games across 6 prominent leagues: the English Premier League, UEFA EuroChamp, Ligue 1 (France), Bundesliga (Germany), Serie A (Italy), and La Liga (Spain). Specifically, 6 matches were recorded at 720p and 50 FPS, 7 at 1080p and 30 FPS, and 7 at 1280p and 30 FPS, as depicted in Fig. 2.

The distribution of the original language of the broadcast game audio is shown in Fig. 3. At the time of submission, a representative subset of the dataset used in this study is publicly available in our GitHub repository: https://github.com/MrFahad/EchoNet.git to support reproducibility, with the full dataset compiled and planned for release upon acceptance. To further ensure the quality of our automatically generated commentary annotations, we first conducted a sampling-based human validation on a stratified subset of 50 clips from the EchoNet++ dataset. These clips were manually verified by native speakers of German, Italian,

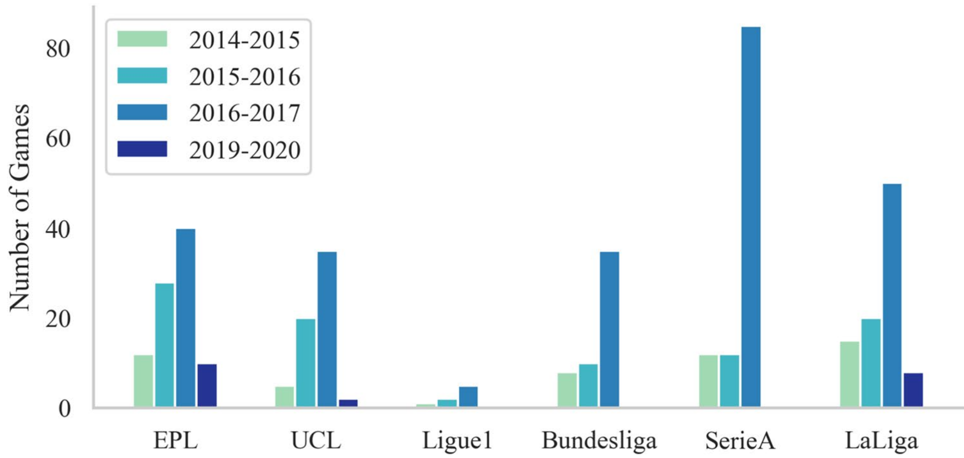

Fig. 2. Overview of the number of games per season consisting of 500 video clips from 20 games.

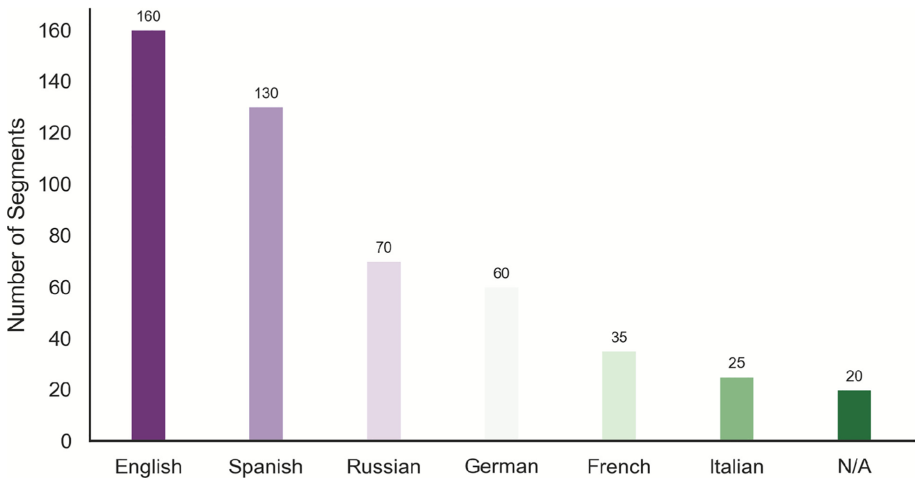

Fig. 3. Distribution of original broadcast audio segments across all 20 games ($\approx$ 1,800 minutes / 30 hours in total), identified through our preprocessing pipeline and analyzed using diverse transcription models. The categories correspond to the detected language in each audio segment, while the "N/A" category represents segments of the videos that contain no audio track.

and English, reflecting the primary languages of commentary in our corpus. Annotators assessed transcript accuracy, segmentation, and noise handling, reporting over 90% agreement with our automatic pipeline.

Following the strong validation results on this subset, we extended human validation to the entire 500 clips dataset, completing a comprehensive manual verification of all annotations. This comprehensive annotation and validation effort ensured the production of high-quality, consistent commentary data across all samples. Overall, the fully curated dataset provides a diverse range of audio-visual environments necessary for evaluating the robustness of our proposed audio processing pipeline across varying production standards, commentary styles, crowd dynamics, and broadcast conditions.

Audio extraction

Before applying any signal processing or speech recognition models, it is crucial to extract high-quality audio from the full-match video recordings. This step ensures that subsequent modules receive a clean and consistent audio representation, irrespective of the input video encoding or format. For this purpose, we use FFmpeg [33], a widely adopted multimedia framework that provides fine-grained control over audio extraction parameters such as sampling rate, number of channels, and encoding format. In our pipeline, we standardize all audio to the following configuration:

Sampling Rate (48 kHz): According to the Nyquist-Shannon sampling theorem [34], to accurately capture all frequencies up to $f_{\max}$, the sampling rate f_s must satisfy that $f_s \geq 2f_{\max}$ for human speech and crowd noise, which typically fall within the 0–20 kHz range, a sampling rate of 48 kHz ensures faithful representation of the signal without aliasing and is a standard used in professional broadcast systems.

Bit Depth (16-bit PCM): Each sample is represented using 16 bits, yielding a dynamic range of DR = $20\log_{10}(2^{16}) \approx 96$ dB. This dynamic range is sufficient to capture both loud crowd sounds and softer commentary without distortion or quantization noise.
Mono Channel: To simplify processing and reduce model complexity, we convert stereo (or multi-channel) audio into a single-channel (mono) signal by averaging the channels as $x_{\text{mono}}(t) = \frac{1}{N}\sum_{i=1}^{N} x_i(t)$ where $x_i(t)$ is the signal from the i-th channel, and N is the number of channels (typically 2 for stereo). This configuration is chosen to maintain a balance between preserving voice fidelity and keeping the computational cost manageable for downstream processing.

```
ffmpeg -i input.mkv -vn -acodec pcm_s16le -ar
    48000 -ac 1 output.wav
```

Listing 1. FFmpeg Audio Extraction Command

This standardization step ensures that all downstream models operate on a uniform representation of the audio, irrespective of the source video's original encoding parameters. The extracted audio is saved in .wav format using 16-bit PCM linear encoding (*pcm_s16le*), which is uncompressed and widely supported for high-fidelity processing.

Preprocessing

To extract semantically meaningful information from raw match audio, we implemented a comprehensive and modular preprocessing pipeline. This pipeline ensures that the audio is first converted into a structured and clean representation before being applied to downstream models, such as speech recognition or speaker classification. The key motivation behind this preprocessing step is to enhance the signal-to-noise ratio (SNR), isolate speech signals, and standardize audio representation across heterogeneous inputs (e.g., different formats, languages, or noise levels). The pipeline comprises five major stages: frequency-domain transformation, filtering, denoising, speech activity detection, and normalization, as described below.

(1) *Frequency-Domain Transformation*: To analyze and visualize the spectral content of the audio, we apply the FFT, which converts a time-domain signal *x*(*t*) into its frequency-domain representation *X*(*f*) using $X(f) = \sum_{n=0}^{N-1} x(n) \cdot e^{-j2\pi fn/N}$ where *x*(*n*) is the discrete time-domain signal, *f* is the frequency index, and *N* is the total number of samples. This transformation enables the identification of dominant frequencies and provides an initial understanding of noise and speech bands. We use the magnitude spectrum |*X*(*f*)| for visualization and analysis.

(2) *Butterworth Bandpass Filter*: To retain only the human voice frequency range, we apply a 5^{th}-order Butterworth bandpass filter with a passband of 300Hz to 7*k*Hz. Butterworth filters are maximally flat in the passband, making them ideal for speech applications where phase distortion and ripples must be minimized.

The frequency response $H(\omega)$ of the Butterworth filter is defined as $|H(\omega)| = \frac{1}{\sqrt{1+\left(\frac{\omega}{\omega_c}\right)^{2n}}}$, where ω_c is the cutoff frequency and *n* is the filter order. This filtering step effectively suppresses low-frequency crowd noise and high-frequency artifacts, ensuring a cleaner and more intelligible speech signal.

(3) *Denoising*: To further improve audio quality, we apply the *Demucs* model, which is based on a convolutional encoder-decoder architecture with bidirectional LSTMs. It separates the input waveform into multiple stems (e.g., vocals, drums, bass). For our task, we isolate the *vocal* stem as it primarily contains speech from commentators and crowd noise. Mathematically, this can be seen as solving a source separation problem where $x(t) = \sum_{i=1}^{M} s_i(t) \quad \Rightarrow \quad \widehat{s}_{\text{vocals}}(t) = \mathcal{D}(x(t))$ where *x*(*t*) is the mixed input, $s_i(t)$ are the source signals, and $\mathcal{D}(\cdot)$ is the Demucs model.

(4) *Speech Activity Detection*: After denoising, the audio is passed through a Speech Activity Detection (SAD) module to identify human speech intervals within the audio timeline. For this, we utilize the Silero VAD (Voice Activity Detection) model, a lightweight neural network classifier trained to distinguish speech from silence and background noise in real-time. The VAD model segments the audio signal $x(t)$ into overlapping windows of fixed duration (e.g., 512 ms) and classifies each window as speech or non-speech and mathematically represented as:

$$\text{VAD}(t) = \begin{cases} 1, & \text{if } \mathbb{P}(\text{speech} \mid x(t)) \geq \theta \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

where $\mathbb{P}(\text{speech} \mid x(t))$ is the posterior probability that speech is active at time*t*, and θ is the confidence threshold (typically set around 0.5–0.7). Once speech regions are detected, the audio is split into smaller segments.

Each segment is then categorized into one of two classes based on the structural characteristics of the audio content:

Commentators. These segments exhibit continuous and structured linguistic features. Their temporal signal exhibits lower spectral entropy and a higher syllable rate, reflecting natural language flow and pauses:

$$H(f) = -\sum_i P(f_i) \log P(f_i) \tag{2}$$

In our pipeline, the spectral entropy is used as a quality assessment measure to identify segments with low speech content or excessive noise, allowing the pipeline to selectively focus on high-quality segments for transcription. This improves overall ASR performance by reducing errors from poor-quality audio without discarding relevant commentary.

Spectators. These are characterized by high-energy bursts (e.g., cheering, booing, clapping) with high temporal and spectral variance and a lack of linguistic content. Spectator noise tends to exhibit flatter frequency distributions (higher entropy) and greater zero-crossing rates:

$$\text{ZCR} = \frac{1}{T-1} \sum_{t=1}^{T-1} \mathbb{I}[x(t) \cdot x(t-1) < 0] \tag{3}$$

where $\mathbb{I}[\cdot]$ is the indicator function and ZCR denotes zero-crossing rate. This classification provides a foundation for applying different processing strategies to structured (speech) and unstructured (ambient) segments, enabling finer-grained downstream analysis.

(5) *Amplitude Normalization*: Finally, we normalize the filtered audio to a fixed amplitude range to reduce loudness variability across recordings. The audio waveform $x(t)$ is normalized, which $\widehat{x}(t) = \frac{x(t)}{\max(|x(t)|)}$ ensures that all subsequent models receive inputs with a standardized dynamic range, improving stability during inference and ensuring that speech intensity does not bias recognition. A multi-stage preprocessing pipeline transforms noisy, variable-quality match audio into clean, structured audio segments suitable for multilingual speech transcription and commentator-spectator separation.

Transcription & translation

Following segmentation and denoising, each speech segment is passed through a series of advanced multilingual automatic speech recognition (ASR) models. This study utilizes four distinct variants of the Whisper architecture, each chosen to explore the trade-offs between computational efficiency, transcription accuracy, and language generalization. These models range from lightweight architectures optimized for rapid inference to large-scale, high-capacity networks that deliver superior accuracy in multilingual contexts.

Table 1 summarizes the core characteristics of the four ASR models integrated into our pipeline, including Whisper (Medium, Large, Turbo) and the optimized IFW. These models process each audio segment independently, enabling parallel evaluation across different model scales and computational trade-offs. The adoption of multiple Whisper variants ensures robustness, flexibility across deployment environments, and scalability for both research and real-time applications. Each model receives an input audio waveform $x(t)$ and outputs a sequence of textual tokens $y = \{y_1, y_2, \ldots, y_T\}$, where each token is predicted as:

$$y_t = \arg\max_{y_t} \mathbb{P}(y_t \mid y_{<t}, x(t)) \tag{4}$$

An encoder-decoder transformer architecture powers this transcription process. The encoder converts the mel-spectrogram $x(t)$ into high-level feature representations, while the decoder employs causal attention to generate the output tokens autoregressively, conditioned on previous outputs.

To maintain the temporal structure of each match and facilitate efficient downstream access, the transcription results are stored in match-specific JSON files 1^{st}`_half.json` for the first half and 2^{nd}`_half.json` for the second half. Each file includes rich metadata such as the segment file name, the raw transcription, the detected language code, and corresponding time intervals.

Model	Params	VRAM Usage	Speed
Whisper(M)	769M	∼5GB	3×
Whisper(L)	1550M	∼10GB	1×
Whisper(Turbo)	1550M	∼6GB	15-20×
IFW	Quantized	∼1.5–2GB	25-30×

Table 1. Comparison of Whisper Models Used in This Study.

Translation and human verification: Given the multilingual nature of soccer commentaries, which may be delivered in Spanish, Russian, French, or other regional languages, we incorporate a translation module to unify downstream textual analysis. For non-English speech segments, we adopt a two-stage strategy: transcription in the source language followed by text-based translation into English using Google Translate [35]. While Whisper supports direct speech-to-English translation, we use this two-step approach to preserve flexibility, transparency, and clearer error analysis across multilingual content.

To ensure transcription fidelity, human annotators validated the original-language transcripts rather than the translations, focusing on segment boundaries, speaker labels, and textual accuracy. Translation quality was quantitatively evaluated using BLEU and Word Correctness Distance (WCD) metrics. We also analyzed the propagation of ASR errors through translation, confirming that robust preprocessing and high-quality ASR substantially reduce translation noise, thereby ensuring reliable multilingual commentary understanding.

Formally, if the detected language of a transcription differs from English, the content is automatically translated using Google Translate. Let $y = \{y_1, \ldots, y_T\}$ represent the sequence of transcribed tokens in the source language L_s, and let $T(\cdot)$ denote the translation function. The translated output y' in the target language L_t = English is given by

$$y' = T(y), \quad \text{where } y' = \{y'_1, \ldots, y'_T\}, \quad L_s \neq L_t \tag{5}$$

This translation step ensures that all transcriptions, regardless of their original language, are normalized to a common linguistic space, facilitating comparison, retrieval, and higher-level reasoning in subsequent stages (see Fig. 4). The translated transcripts are stored in structured JSON files named `translated_1`st`_half.json` and `translated_2`nd`_half.json`, corresponding to the two halves of each match. These files retain the structure and timestamps of the original transcription JSONs while introducing additional fields that capture the translated text and the automatically detected source language.


```
[
  {
    "file": "spectator_clip1.wav",
    "type": "spectator",
    "lang": "de",
    "original": "1-1 wäre für Darmstadt auch ein Traumergebnis.",
    "translated": "1-1 would also be a dream result for Darmstadt."
  },
  {
    "file": "commentator_clip_0.wav",
    "type": "commentator",
    "lang": "de",
    "original": "Ich fand, überragenden Spiel gegen Leverkusen. Das ist auf ganz, ganz wenige
Ausnahme. Da muss man den Haar schon ein paar Mal spalten.",
    "translated": "I found outstanding game against Leverkusen.This is very few exception.You
have to split your hair a few times."
  },
]
[
  {
    "file": "spectator_clip7.wav",
    "type": "spectator",
    "lang": "es",
    "original": "Consigue despejar, buen control de De Bruyne, el pase es para Stern.",
    "translated": "Get clear, good control of De Bruyne, the pass is for Stern."
  },
  {
    "file": "commentator_clip4.wav",
    "type": "commentator",
    "lang": "es",
    "original": "le gusta, con lo que más a gusto se siente, con esos balones portales.",
    "translated": "He likes, with what he feels most comfortable, with those portal balls."
  },
]
[
  {
    "file": "spectator_clip1.wav",
    "type": "spectator",
    "lang": "en",
    "original": "Might still fall for the sweep but it's away by Ken Carver.",
    "translated": "Might still fall for the sweep but it's away by Ken Carver."
  },
  {
    "file": "commentator_clip2.wav",
    "type": "commentator",
    "lang": "en",
    "original": "That will buy round some time, only Grige-en-Qui is up there.",
    "translated": "That will buy round some time, only Grige-en-Qui is up there."
  },
]
```


Fig. 4. Multilingual commentary examples from three major soccer leagues. Whisper transcriptions and English translations are shown for original audio commentary in German (Bundesliga), Spanish (La Liga), and English (EPL), illustrating our EchoNet pipeline performance across languages.

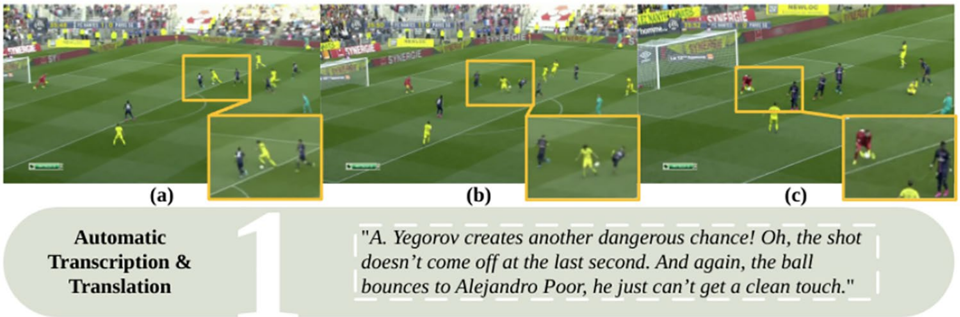

Fig. 5. Frame (a) to (c) transcription using Whisper and English translation of Russian commentary from Ligue 1 match, with key frames illustrating corresponding segments.

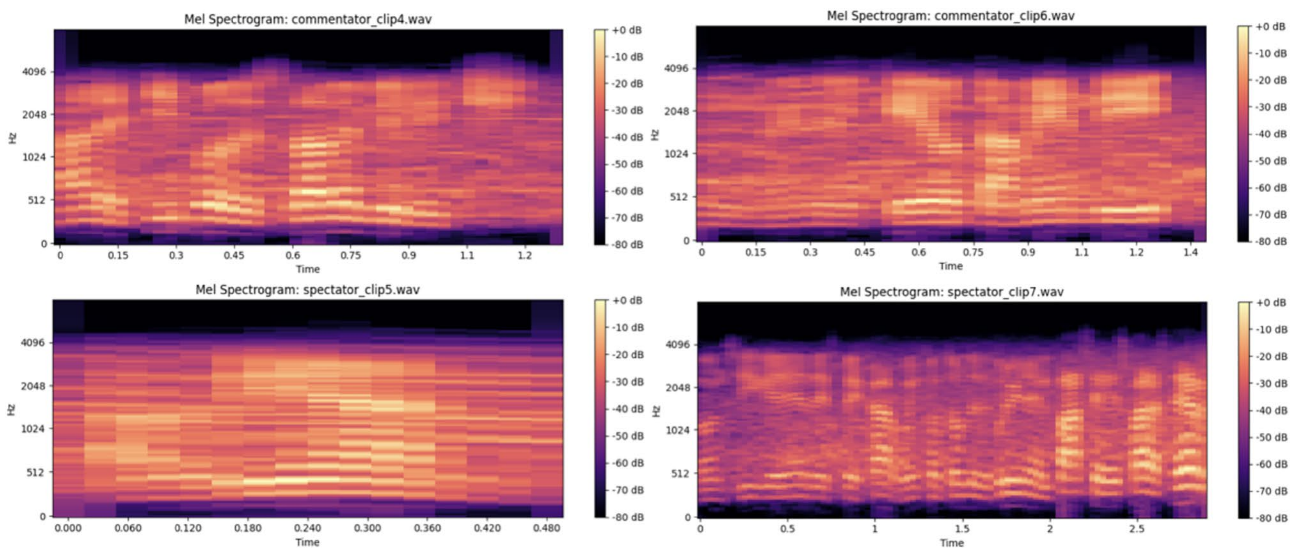

Fig. 6. The top-row shows the Mel Spectrogram of the Commentator, with the same time-frequency and dB scale. While the bottom-row shows the Mel spectrogram of the Spectator showing time vs. frequency (Hz) with intensity in decibels (dB).

This end-to-end sequence from voice activity segmentation to multilingual transcription, human verification, and translation enables a scalable, language-independent framework for processing soccer match audio. The resulting data is enriched with meaningful metadata, facilitating efficient analysis, retrieval, and integration into downstream tasks. Fig. 5 presents a visual overview of multilingual transcription and translation from a Ligue 1 match using Whisper. A discussion of this design choice and its trade-offs with respect to direct speech-to-English translation is provided in Sec. 5.1.

Frame-level feature analysis

In our pipeline, the initial FFT during preprocessing serves to isolate and enhance relevant frequency components, effectively reducing noise and standardizing the audio signal. Subsequently, a second FFT is applied during feature extraction to generate a magnitude spectrum, which is then transformed into a Mel spectrogram. This time-frequency representation aligns with human auditory perception. For multimodal analysis, we extract the top 50 frames from the video corresponding to the audio segments, ensuring that both audio and visual data are temporally synchronized for subsequent transcription and analysis. As shown in Fig. 6, the Mel spectrograms of the commentator and spectator reveal distinct patterns in the frequency distributions.

Results

This section presents a comprehensive evaluation of the proposed EchoNet pipeline on the EchoNet++ dataset. We report both task-level and signal-level results to assess transcription accuracy, linguistic fidelity, and perceptual audio quality. Task-level metrics, including Word Error Rate (WER), Character Error Rate (CER), BLEU score, and Word Count Distribution (WCD), quantify the effectiveness of our automatic speech recognition (ASR) and translation components. Similarly, signal-level measures such as Perceptual Evaluation of Speech Quality (PESQ), Short-Time Objective Intelligibility (STOI), Scale-Invariant Signal-to-Distortion Ratio

(SI-SDR), and Log-Spectral Distance (LSD) evaluate the clarity and intelligibility of the extracted commentary signals. Together, these results provide a holistic assessment of both the linguistic and acoustic performance of EchoNet++ under realistic match conditions.

Evaluation metrics

To quantitatively evaluate the performance of the proposed EchoNet pipeline on the EchoNet++ dataset, we adopt a set of complementary task-level and signal-level metrics that jointly assess transcription accuracy, linguistic fluency, and perceptual audio quality.

Task-level metrics: Automatic speech recognition (ASR) and translation performance are measured using Word Error Rate (WER) and Character Error Rate (CER), which capture insertion, deletion, and substitution errors at the word and character levels, respectively. The BLEU score further evaluates *n*-gram fluency and lexical coherence between the predicted and reference transcripts, while the Word Count Distribution (WCD) examines the coverage and frequency of domain-specific vocabulary, including player names, tactical terms, and match events. Together, these metrics provide a detailed characterization of linguistic accuracy and contextual fidelity in the generated transcripts.

Signal-level metrics: To assess the perceptual and acoustic quality of extracted commentary signals, we report Perceptual Evaluation of Speech Quality (PESQ) and Short-Time Objective Intelligibility (STOI), which measure perceived clarity and intelligibility under stadium noise conditions. Additionally, the Scale-Invariant Signal-to-Distortion Ratio (SI-SDR) quantifies separation accuracy between commentary and background audio, while the Log-Spectral Distance (LSD) captures fine-grained spectral preservation. These measures collectively reflect the system's ability to recover clean, intelligible, and natural-sounding commentary from complex audio environments.

Report accuracy (RA): To further quantify the individual contribution of each module within the EchoNet pipeline, we introduce the Report Accuracy (RA) metric as part of a component-level ablation study on the EchoNet++ dataset. RA measures the degree to which each subsystem, such as speech enhancement, ASR, translation, and alignment, contributes to the correctness of the final commentary reports. RA is defined as an interpretable proxy of transcription reliability derived from the Word Error Rate (WER). Specifically, we estimate RA as

$$\mathrm{RA} \approx 100 - \alpha \cdot \mathrm{WER}, \tag{6}$$

where $\alpha = 0.95$ is a scaling factor selected empirically to align RA values with the perceived correctness of commentary segments in downstream reporting tasks. This scaling reflects the observation that not all word-level errors affect the semantic correctness or usability of match reports equally, particularly in noisy broadcast conditions. Importantly, RA is not intended to replace standard ASR metrics such as WER or CER, but rather to provide a single, intuitive score that summarizes practical report-level correctness and facilitates comparison of module-level contributions within the pipeline.

Implementation details

The complete EchoNet pipeline was implemented in `Python 3.12.7` using `PyTorch 2.7.1` and `librosa` for signal processing. All experiments were executed on an NVIDIA `RTX 4080` GPU (16 GB). The pipeline operates in three main stages. In the audio preprocessing stage, raw video input is first decoded to extract the audio stream $A(t)$, followed by a short-time Fourier transform (STFT) and bandpass filtering between 300 Hz and 7 kHz to suppress low-frequency crowd noise and high-frequency artifacts. Denoising is performed using the pre-trained `Demucs` model, and speech segments are identified through a Silero VAD algorithm. Each detected segment is then normalized by its maximum amplitude to obtain $A_{\mathrm{norm}}(t)$. In the transcription and translation stage, the normalized speech segments are transcribed using `Whisper`, yielding transcripts T_i. A language detection module identifies the commentary language ℓ, and non-English transcriptions are translated into English using the google translator. The final transcript T_{final} is selected as T_i if $\ell = \mathrm{en}$, or its translated version otherwise. The feature and frame analysis stage extracts FFT and Mel-spectrogram features from $A_s(t)$, followed by top-50 salient frame sampling based on speech energy and visual motion cues. The resulting annotations, timestamps, and features are stored in structured `CSV` and `JSON` formats for downstream analysis. All modules were integrated within a unified pipeline, ensuring synchronized audio–visual alignment and full reproducibility.

Quantitative results

Task-level evaluation: To quantitatively assess the task-level multilingual transcription and translation performance of our EchoNet pipeline, we evaluate transcription and translation quality across two benchmark datasets and on our curated EchoNet++ dataset. The GOAL [32] dataset comprises human-verified transcriptions from 40 half-time game videos across 20 games, with original English commentary from the SoccerNet dataset. The SoccerNet-Echo dataset comprises of 550 games from 6 different leagues and 4 different seasons. Table 2 presents a comparative performance analysis of our pipeline across multiple automatic speech recognition (ASR) and translation quality metrics. Fig. 8 quantifies the improvements achieved after preprocessing across both task-level (WER) and signal-level metrics (PESQ, STOI, SI-SDR, and LSD). The results demonstrate consistent reductions in WER and complementary gains in perceptual and intelligibility measures across both benchmark datasets and on our EchoNet++ dataset. This joint analysis highlights that the preprocessing pipeline not only improves transcription accuracy but also enhances overall signal quality under diverse acoustic conditions.

Dataset	Model	WER↓	CER↓	BLEU↑	WCD↑
Before preprocessing (Raw audio input)					
GOAL	Whisper(M)	17.8	11.2	0.58	12.2K
	Whisper(L)	16.7	10.8	0.59	12.7K
	Whisper(Turbo)	16.0	10.1	0.60	13.0K
	IFW	15.1	9.5	0.63	13.1K
SoccerNet-Echo	Whisper(M)	19.2	12.4	0.55	11.3K
	Whisper(L)	18.0	11.6	0.57	11.6K
	Whisper(Turbo)	17.6	11.3	0.58	12.0K
	IFW	16.5	10.7	0.61	12.2K
EchoNet++ (Ours)	Whisper(M)	20.4	13.0	0.51	10.5K
	Whisper(L)	19.1	12.2	0.52	11.0K
	Whisper(Turbo)	18.7	11.9	0.53	11.4K
	IFW	17.6	11.1	0.57	11.7K
After Preprocessing (Denoising, VAD, Filtering)					
GOAL	Whisper(M)	12.5	7.4	0.68	13.2K
	Whisper(L)	11.4	7.2	0.69	13.5K
	Whisper(Turbo)	10.1	6.5	0.71	14.0K
	IFW	**9.3**	**6.1**	**0.75**	**14.3K**
SoccerNet-Echo	Whisper(M)	13.8	8.2	0.65	12.5K
	Whisper(L)	12.5	8.0	0.66	13.0K
	Whisper(Turbo)	11.6	7.3	0.69	13.6K
	IFW	**10.5**	**6.7**	**0.73**	**13.7K**
EchoNet++ (Ours)	Whisper(M)	14.1	8.5	0.63	11.9K
	Whisper(L)	13.1	8.3	0.64	12.3K
	Whisper(Turbo)	12.4	7.7	0.67	12.9K
	IFW	**11.2**	**7.1**	**0.70**	**13.3K**

Table 2. Task-level performance evaluation before and after applying our preprocessing pipeline (EchoNet) across both benchmark datasets and on our EchoNet++ dataset using Whisper models. Significant values are in bold.

Model	Before	After	Improvement
Whisper(M)	17.9	13.7	4.2
Whisper(L)	16.8	12.9	3.9
Whisper(Turbo)	16.2	11.7	4.5
IFW	18.5	14.2	4.3

Table 3. Impact of denoising on ASR transcription accuracy WER accross all models on the EchoNet++ dataset.

Dataset	PESQ ↑	STOI ↑	SI-SDR (dB)↑	LSD↓
Before preprocessing				
GOAL	2.10	0.81	10.3	2.03
SoccerNet-Echo	2.05	0.79	9.8	2.14
EchoNet++ (Ours)	2.28	0.84	11.7	1.76
After preprocessing				
GOAL	3.12	0.93	15.8	1.23
SoccerNet-Echo	3.15	0.91	15.2	1.28
EchoNet++ (Ours)	3.21	0.93	16.3	1.19

Table 4. Signal-level performance evaluation before and after applying our preprocessing pipeline (EchoNet) across both benchmark datasets and on our EchoNet++ dataset using Whisper models.

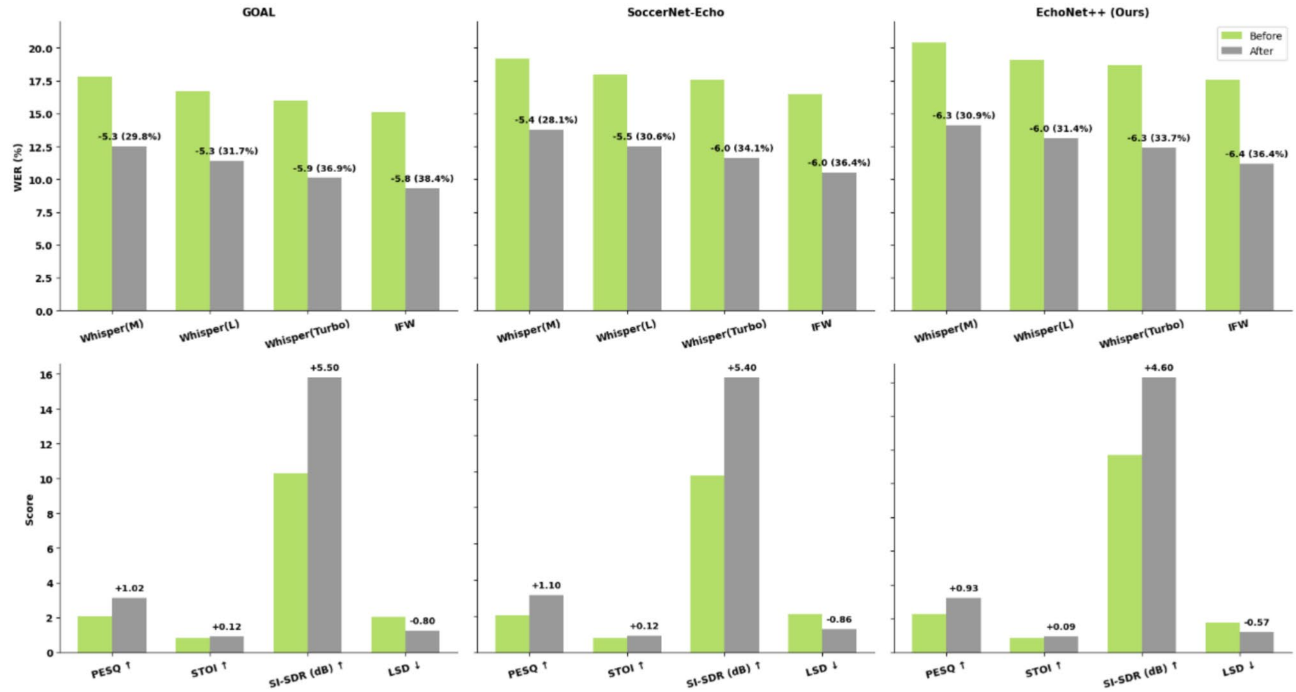

Fig. 7. Quantitative comparison of speech enhancement performance across both benchmark datasets and on our EchoNet++ dataset. Top row: Task-level metrics WER (↓) across four ASR models (Medium, Large, Turbo, and IFW) before and after enhancement. Relative improvements in WER are annotated above the bars. Bottom row: Signal-level metrics PESQ (↑), STOI (↑), SI-SDR (↑), and LSD (↓) before and after enhancement. Improvements in signal metrics are annotated above the bars.

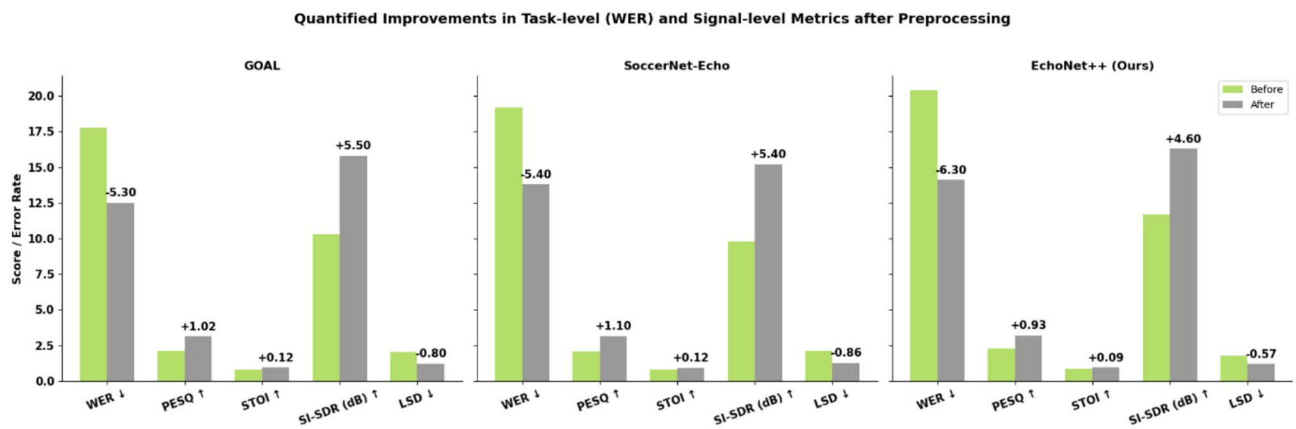

Fig. 8. Task-level and signal-level performance improvements across both benchmark datasets and on our EchoNet++ dataset before and after applying our preprocessing pipeline. Each bar pair reports absolute scores for five evaluation metrics: WER (↓), PESQ (↑), STOI (↑), SI-SDR in dB (↑), and LSD (↓). Numerical annotations above bars indicate the magnitude of improvement, where negative values denote error reduction (WER, LSD) and positive values denote quality gains (PESQ, STOI, SI-SDR).

To evaluate the impact of demucs on transcription quality, we measured WER on the EchoNet++ dataset before and after denoising. Table 3 summarizes the results, showing that denoising consistently improves transcription accuracy across all models, confirming its benefit as a preprocessing step for robust audio processing.

Signal-level evaluation: To assess the effectiveness of our audio preprocessing pipeline, we evaluate signal-level performance on three datasets: GOAL, SoccerNet-Echo, and our proposed EchoNet++. We report four key metrics: Perceptual Evaluation of Speech Quality (PESQ) [36], Short-Time Objective Intelligibility (STOI) [37], Scale-Invariant Signal-to-Distortion Ratio in decibels (SI-SDR), and Log-Spectral Distance (LSD) [38]. These metrics respectively quantify perceptual speech quality, intelligibility, distortion robustness, and spectral fidelity. Results demonstrate significant improvements across all metrics after applying EchoNet's pipeline, confirming

Dataset	Model	Before	After
GOAL	Whisper(M)	83.1	88.1
	Whisper(L)	84.1	89.2
	Whisper (Turbo)	84.8	90.4
	IFW	**85.6**	**91.2**
SoccerNet-Echo	Whisper(M)	81.8	86.9
	Whisper(L)	82.9	88.1
	Whisper (Turbo)	83.3	88.9
	IFW	**84.3**	**90.0**
EchoNet++ (Ours)	Whisper(M)	80.6	86.6
	Whisper(L)	81.9	87.6
	Whisper (Turbo)	82.3	88.2
	IFW	**83.3**	**89.4**

Table 5. Report Accuracy (RA) performance evaluation before and after applying our preprocessing pipeline (EchoNet) across benchmark datasets and the EchoNet++ dataset. RA is an interpretable, task-level proxy derived from WER and is estimated as $\mathrm{RA} \approx 100 - 0.95 \cdot \mathrm{WER}$, where the scaling factor is chosen to better reflect report-level usability. Significant values are in bold.

Dataset	Model	WER	RA	BLEU / WCD
SoccerNet-Echoes	Whisper(Large-v1)	44.3	58.0	54.5 / 0.261
	Whisper(Large-v2)	45.8	56.6	52.6 / 0.269
	Whisper(Large-v3)	55.1	47.6	48.0 / 0.341
	EchoNet (IFW)	**10.0**	**90.0**	**92.3** / **0.072**

Table 6. System-level evaluation of the EchoNet pipeline on SoccerNet-Echoes. RA is estimated as $100 - 0.95 \cdot \mathrm{WER}$. BLEU and WCD quantify translation quality. Whisper Large variants are baseline models; EchoNet (IFW) represents our full preprocessing pipeline. Significant values are in bold.

its robustness under real-world multilingual broadcast noise and overlap scenarios. Table 4 summarizes the comparative results.

To provide a holistic view of enhancement effectiveness, we further compare both task-level and signal-level performance across benchmark datasets and our EchoNet++ dataset. Fig. 7 presents this quantitative comparison, integrating automatic speech recognition (ASR) performance and perceptual quality metrics. The to p panel illustrates task-level outcomes in terms of Word Error Rate (WER) across multiple ASR models, while the bottom panel reports corresponding signal-level metrics (PESQ, STOI, SI-SDR, and LSD), highlighting consistent gains after enhancement. Fig. 8 further quantifies these improvements before and after preprocessing across all datasets. The results demonstrate consistent reductions in WER and complementary gains in perceptual and intelligibility measures, confirming that the preprocessing pipeline not only improves transcription accuracy but also enhances overall signal quality under diverse acoustic conditions. Together, these results validate that improvements at the signal level directly translate to enhanced downstream ASR performance.

Report accuracy evaluation: To assess the practical transcription reliability of our EchoNet pipeline, we employ the Report Accuracy (RA) metric across the GOAL, SoccerNet-Echo, and our curated EchoNet++ datasets. RA provides an intuitive, task-level measure of the proportion of correctly recognized content, estimated as RA $\approx 100 - 0.95\cdot$ WER, directly linking it to word-level transcription errors. The scaling factor was chosen based on preliminary analysis to ensure that RA better reflects report-level usability rather than raw word-level error counts.

Table 5 presents RA values before and after applying our preprocessing pipeline, which incorporates denoising, voice activity detection (VAD), and spectral filtering. The results show that our preprocessing consistently improves RA across all datasets, highlighting the effectiveness of EchoNet in enhancing transcription fidelity under diverse stadium audio conditions. While RA is primarily a content-based metric, improvements in signal-level quality (e.g., PESQ, STOI, SI-SDR, LSD) can indirectly boost RA by making the audio more intelligible and easier for ASR models to transcribe accurately. This metric complements standard WER, CER, BLEU, and WCD measures by providing a single, interpretable score for overall report correctness and the practical utility of generated transcripts in downstream applications such as match summarization and analytics.

System-level evaluation: To validate the practical utility of the EchoNet pipeline, we applied it to the SoccerNet-Echoes dataset and compared the generated annotations with the official transcripts. As shown in Table 6, our preprocessing pipeline, combined with the IFW model, consistently improves ASR metrics (WER, RA) and translation quality (BLEU, WCD) over the baseline Whisper Large variants. These results demonstrate that the generated annotations are high-fidelity, directly reusable, and robust for downstream tasks such as

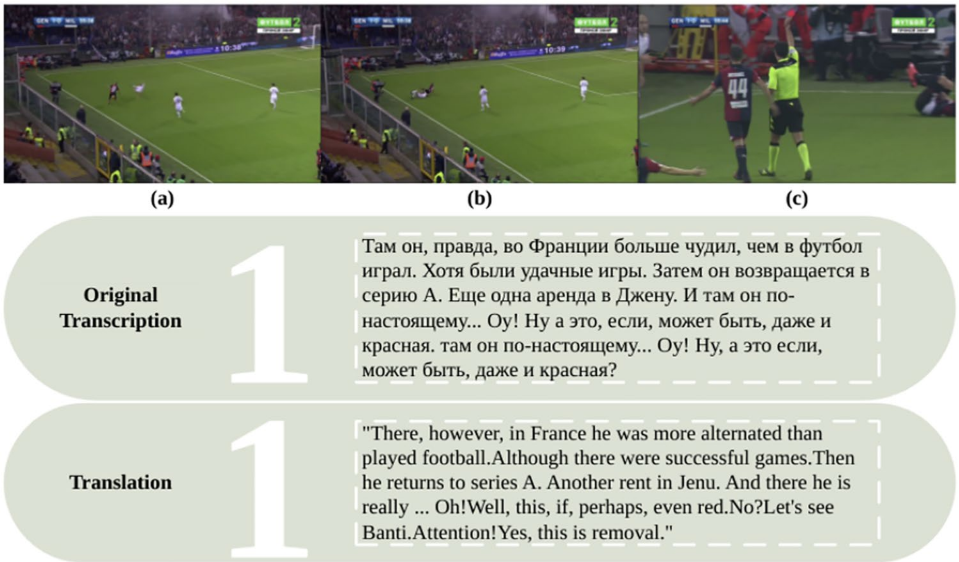

Fig. 9. Frame (**a**) to (**c**) transcription using Whisper and English translation of the Russian commentary from a Serie A match, with key frames illustrating corresponding segments.

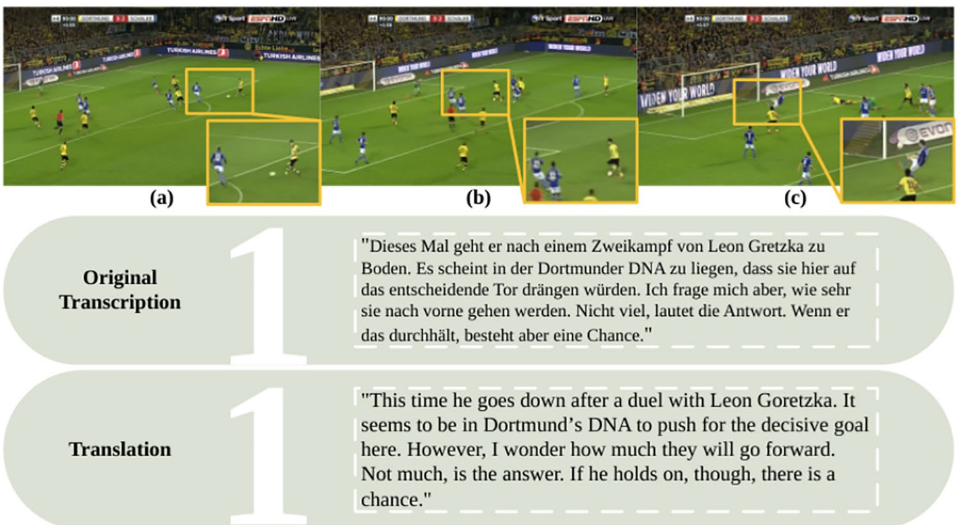

Fig. 10. Frame (**a**) to (**c**) transcription using Whisper and English translation of the German commentary audio from a Bundesliga match, with key frames illustrating corresponding segments.

summarization, cross-lingual commentary analysis, and sports analytics, thereby highlighting the practical value and reproducibility of our approach.

Qualitative evaluation

Beyond quantitative metrics, we conduct a qualitative evaluation to gain deeper insight into the linguistic fidelity and contextual accuracy of the ASR outputs. This includes an analysis of vocabulary richness, handling of domain-specific terminology, sentence fluency, and the models' ability to preserve semantic integrity in noisy and multilingual environments. Furthermore, we justify the selection of each ASR model by examining its linguistic coverage, consistency across broadcast conditions, and alignment with the demands of real-world soccer commentary. This evaluation reveals nuanced differences in model performance that are not captured by error rates alone. Fig. 9 presents a visual overview of multilingual transcription from a Serie A and its English translation from the original game audio in Russian, while Fig. 10 shows a multilingual transcription example from a Bundesliga and its English translation from the original game audio in German using Whisper, respectively.

Config.	WER↓	SI-SDR↑	RA	Time
w/o Bandpass + IFW	18.4	9.7	75.1	13.2
w/o Demucs + IFW	16.3	N/A	77.2	12.7
w/o VAD + IFW	12.5	11.8	81.4	14.1
w/o Normalization + IFW	9.2	12.9	85.6	15.0
Whisper(L) (Full)	8.2	13.2	87.5	11.1
Whisper(M) (Full)	11.9	13.2	80.4	8.6
Combined	6.1	13.2	90.3	15.5

Table 7. Component-level ablation results on the EchoNet++.

Pipeline Variant	WER↓	CER↓	BLEU↑	RA
Only Denoising	13.7	8.3	0.64	86.8
Only VAD	14.2	8.8	0.63	86.1
Only Filtering	13.8	8.4	0.64	86.6
Combined	12.4	7.7	0.67	88.2

Table 8. Ablation of preprocessing stages on EchoNet++ dataset using Whisper(Turbo). RA $\approx 100 - 0.95\cdot$ WER.

Ablation study

To quantify the individual contribution of each pipeline module, we perform a component-level ablation study on the EchoNet++ dataset. Ablations includes, removing Bandpass, Demucs, VAD, or Normalization, and comparing Whisper (Medium, Large and IFW) with the full pipeline. Evaluation metrics were WER, SDR, Report Accuracy (RA), and inference time; see Table 7 below.

Findings: Bandpass filtering was critical, improving SDR by 3–4 dB and reducing WER from 18.4% to 6.1%. Demucs separation further cut WER by 10% (goal calls vs. crowd noise). VAD reduced non-speech hallucinations, improving RA from 81.4% to 90.3%. Normalization stabilized amplitudes, lowering WER to 9.2%. The IFW model excelled in capturing soccer jargon (IFW: WER 6.1%, RA 90.3%) with higher latency, while Whisper(M) showed lower accuracy but faster performance.

Error analysis: Bandpass aided $\sim$50% noisy clips with 10–15 dB SNR gains; Demucs improved 45% by separating cheers vs. speech; VAD prevented silence-induced errors (30%); Normalization avoided clipping in 20%. Whisper(M,L) often misrecognized jargon (e.g., “offside”), while IFW occasionally misaligned fast speech ($\sim$12% of clips). In $\sim$8% low-quality audio, RA dropped $<$ 80%. Overall, each module is essential for robust commentary transcription.

Preprocessing ablation: We first analyze the contribution of individual preprocessing stages, including denoising, voice activity detection (VAD), and spectral filtering. Table 8 shows the effect of applying only one stage at a time, highlighting that each module contributes positively to transcription performance. Denoising has the most significant impact on Report Accuracy (RA), followed by VAD and spectral filtering.

We further analyzed the impact of the preprocessing sequence and parameter choices on transcription performance. Swapping the order of denoising, VAD, and filtering results in slight decreases in RA (up to 1.5%), confirming that the chosen sequence (denoise $\rightarrow$ VAD $\rightarrow$ filter) is optimal. Similarly, moderate variations in VAD thresholds and filter bandwidths cause only minor changes in RA ($0.5 - -1.0\%$), demonstrating that our pipeline is robust to reasonable parameter adjustments while maintaining high transcription fidelity.

Discussion

EchoNet++ is constructed from our curated SoccerPro dataset to support research in automatic speech recognition, natural language processing, and multimodal learning for sports analytics. We emphasize that the goal of this work is not dataset scale alone, but the principled evaluation of end-to-end signal processing decisions in challenging broadcast environments. Our analysis demonstrates that the EchoNet pipeline, which includes denoising, voice activity detection, and filtering, consistently improves transcription accuracy and report reliability across GOAL, SoccerNet-Echo, and EchoNet++ datasets. Improvements in signal-level quality metrics (PESQ, STOI, SI-SDR, and LSD) contribute to increased intelligibility, which in turn enhances Report Accuracy (RA) and overall transcription fidelity. These results highlight the practical utility of our derived resources, like denoised audio, transcripts, and annotations for downstream tasks such as match summarization and analytics. While the dataset originates from broadcast recordings, speech is limited to professional commentators and referees, minimizing privacy concerns. Access to EchoNet++ is controlled under a research-only license, ensuring ethical and legal compliance. Applying EchoNet to the SoccerNet-Echoes dataset demonstrates that our preprocessing pipeline generalizes beyond our curated EchoNet++ dataset. The resulting annotations show improved transcription fidelity and translation quality, confirming that our approach produces high-quality, reusable data suitable for summarization, cross-lingual analysis, and other downstream analytics. Overall, our findings show that preprocessing combined with a carefully curated dataset enhances both transcription quality and signal integrity, providing a reliable benchmark for reproducible research in multimodal sports analytics.

Design choices and limitations

While Whisper provides a direct speech-to-English translation mode, we adopt a two-stage approach consisting of source-language transcription followed by text-based translation using Google Translate. This design prioritises modularity and interpretability, enabling the independent analysis of transcription and translation errors. Although a direct translation approach may reduce pipeline complexity, the chosen strategy provides greater flexibility when working with multilingual broadcast data and heterogeneous language distributions.

Applications

The proposed pipeline enables multiple downstream tasks by leveraging rich transcriptions and structured audio streams. Below, we highlight key applications demonstrating its practical utility in soccer media analytics.

Commentary understanding: The transcribed commentator speech enables automatic identification of tactical events (e.g., goals, fouls, saves) and emotional moments through keyword spotting and sentiment analysis. This supports applications like highlight generation and tactical annotation.

Audience response analysis: Spectator audio is analyzed to detect peaks in crowd reactions, revealing key moments of excitement. The system also identifies chants and national anthems, facilitating sociocultural insights and enhancing fan-centric broadcasting.

Game summarization: By aligning commentator cues and crowd responses, the framework generates concise match summaries enriched with both tactical and emotional context, supporting multilingual storytelling and media automation.

Conclusion

We presented EchoNet, a comprehensive, scalable, and reproducible pipeline for processing and analyzing professional soccer match audio from broadcast videos. Our multi-stage framework integrates signal processing, deep-learning-based denoising, voice activity detection, multilingual transcription using Whisper and IFW models, and text translation, producing temporally aligned, semantically rich, and language-agnostic outputs. The system reliably distinguishes structured commentator speech from crowd noise, enabling robust downstream applications such as commentary analysis, match summarization, cross-lingual sentiment tracking, and audio-based event retrieval. Through systematic component- and task-level evaluations on EchoNet++, GOAL, and SoccerNet-Echoes, we demonstrated that preprocessing, segmentation, and transcription choices collectively improve both transcription fidelity and translation quality. By releasing a curated subset of EchoNet++ and detailed pipeline documentation, we provide a benchmark for reproducible research in audio-centric sports analytics.

Looking ahead, we plan to extend the pipeline with speaker diarization to distinguish multiple commentators, event detection aligned with match moments, and crowd behavior modeling leveraging audio-visual cues. These enhancements will further expand EchoNet's capabilities, providing a versatile framework for advanced multimedia and sports analytics research.

Data availability

A subset of the dataset used in this study is available on GitHub (https://github.com/MrFahad/EchoNet.git); the full dataset and processing pipeline will be released in a structured format.

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Author contributions

Fahad Majeed served as the lead author and was primarily responsible for the conceptualization and execution of the research. Jens Schneider contributed to the manuscript through thorough review and technical writing. Marco Agus and Maria Nazir also provided critical review and technical input, supporting the development and refinement of the work.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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